

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 02****Topics covered :**

1. Explicit Solution and Implicit Solution of a differential equation
2. General solution, particular solution and singular Solutions

- A differential equation involving derivatives of a function of a single variable is called the **ordinary differential equation (ODE)**. Some examples of ODE are [1,2]

1. Let $y(t)$ denote the vertical position (measured from the ground with the upward direction as positive) of an object (of mass m) under free fall with linear air drag. According to Newton's second law,

$$\frac{d^2y}{dt^2} = -g - \left(\frac{k}{m}\right) \frac{dy}{dt} \quad (1)$$

where g is the acceleration due to gravity and k is the constant related to air resistance.

2. According to Newton's law of cooling, the rate of change of temperature $T(t)$ of an object is proportional to the difference between $T(t)$ and the temperature of the surrounding medium T_s . Mathematically,

$$\frac{dT}{dt} = -k(T - T_s) \quad (2)$$

where k is a positive constant.

3. The current response $i(t)$ of a series RLC circuit driven by a time-varying voltage source $v(t)$ is given by

$$L \frac{d^2i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i(t) = \frac{dv(t)}{dt} \quad (3)$$

where R , L and C denote the resistance, inductance and capacitance respectively.

4. In electromagnetics, the Legendre's equation arises while solving the Laplace equation in spherical coordinates. The Legendre's equation is given by

$$(1 - x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + l(l + 1)y = 0 \quad (4)$$

where l is a constant.

5. The Bessel's equation arises while solving the Laplace equation in cylindrical coordinates. The Bessel's equation is given by

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - p^2)y = 0 \quad (5)$$

where p is a constant.

- A differential equation that involves partial derivatives of a function of multiple variables is called a [partial differential equation \(PDE\)](#). Some examples of PDE are [1, 2]

1. The Laplace equation, which appears in various fields, especially in electromagnetics is a PDE given by

$$\frac{\partial \phi}{\partial x^2} + \frac{\partial \phi}{\partial y^2} + \frac{\partial \phi}{\partial z^2} = 0 \text{ (often written as) } \nabla^2 \phi(x, y, z) = 0 \quad (6)$$

where $\phi(x, y, z)$ is the potential function and ∇^2 is the Laplacian operator.

2. The Poisson's equation that describes the potential due to a charge distribution is a PDE given by

$$\nabla^2 \phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon} \quad (7)$$

or, in Cartesian coordinates,

$$\frac{\partial \phi}{\partial x^2} + \frac{\partial \phi}{\partial y^2} + \frac{\partial \phi}{\partial z^2} = g(x, y, z) \quad (8)$$

where $\phi(x, y, z)$ is the potential function, ρ is the charge density and ϵ is the permittivity. The general form of the Poisson's equation is

$$\nabla^2 \phi(\mathbf{x}) = g(\mathbf{x}) \quad (9)$$

3. The wave equation that describes the propagation of electromagnetic waves (or any other waves) through a medium is a PDE given by

$$\frac{\partial^2 \phi(\mathbf{r}, t)}{\partial t^2} = \omega^2 \nabla^2 \phi(\mathbf{r}, t) \quad (10)$$

- The [order](#) of a differential equation is the order of the highest derivative in the equation. For example, (2) is a first order ODE while (1), (3)-(5) are second order ODEs. The heat equation ($\frac{\partial \phi}{\partial t} = c^2 \nabla^2 \phi$), Laplace equation (6), Poisson's equation (9), and the wave equation (10) are examples of second order PDEs.

The [degree](#) of the differential equation is the power to which the highest derivative is raised (unless the differential equation involves fractional derivatives).

- The [general form](#) of an ODE of order n is

$$F(x, y, y', \dots, y^{(n)}) = 0 \quad (11)$$

where F is a real valued function of $n + 2$ variables. On the other hand, the [normal form](#) an ODE of order n is

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)}) \quad (12)$$

In the context of first-order ODE, the general form is often considered as

$$y' = -\frac{P(x, y)}{Q(x, y)} \quad (13)$$

- The function $u = f(x)$ is a [solution](#) (or [explicit solution](#), to be more precise) to (11) on the interval I , if $u', \dots, u^{(n)}$

exist and $F(x, u, u', \dots, u^{(n)}) = 0 \forall x \in I$.

Examples:

1. The function $y(t) = kt + c$, where c is a constant, is a solution to the differential equation $\frac{dy}{dt} = k$.
 2. The function $y(t) = ce^{kt}$, where c is a constant, is a solution to the ODE $\frac{dy}{dt} = ky$.
 3. The function $\phi(x, t) = Ae^{jc(x-\omega t)}$, where A and c are constants, satisfies the one-dimensional wave equation (which is a PDE) $\frac{\partial^2 \phi}{\partial t^2} = \omega^2 \frac{\partial^2 \phi}{\partial x^2}$. Although $\phi(x, t) = Ae^{jc(x-\omega t)}$ satisfies the wave equation, it does not represent a solution in the classical sense, as it is complex-valued, whereas solutions to the wave equation are typically required to be real-valued. Therefore, the solution to the one-dimensional wave equation is given by $\phi(x, t) = A \cos(c(x - \omega t))$.
- The relation $u(x, y) = 0$ is an **implicit solution** [3] of (11) on an interval I , if
 - a) $u(x, y)$ defines y as an implicit function of x on I , i.e., if there exists $g(x)$ on I such that $u(x, g(x)) = 0 \forall x \in I$, and
 - b) $F(x, g(x), g'(x), \dots, g^{(n)}(x)) = 0 \forall x \in I$

Example:

1. The relation $x^2 + y^2 = c^2$ is an implicit solution to the differential equation $\frac{dy}{dx} = -\frac{x}{y}$ in the interval $-c < I < c$ (**Determine why the relation is not a solution in the interval $-c \leq I \leq c$?**). Note that the differential equation when put in the general form is given by $F(x, y, y') = yy' + x = 0$ and is defined for $x \in \mathbb{R}$. Note that in this case, the choice of $g(x) = \pm\sqrt{c^2 - x^2}$.
- Consider the first-order differential equation

$$\frac{dy}{dx} = f(x) \tag{14}$$

The solution to (14) is given by

$$y = f_1(x) = \int f(x)dx + c_1 \tag{15}$$

where c_1 is an arbitrary constant. Similarly, for the second-order differential equation

$$\frac{d^2y}{dx^2} = f(x) \tag{16}$$

the solution $f_2(x)$ can be obtained by integrating both sides of (15), yielding

$$y = f_2(x) = \int f_1(x)dx + c_1x + c_2 \tag{17}$$

where c_1 and c_2 are arbitrary constants.

By extending this method, it can be shown that the solution to a general n -th order differential equation:

$$\frac{d^ny}{dx^n} = f(x) \tag{18}$$

is given by

$$y = f_n(x) = \int f_{n-1}(x)dx + c_1x^{n-1} + \cdots + c_n \quad (19)$$

where $c_1, \dots, c_n$ are arbitrary constants.

The preceding analysis leads to two important conclusions.

1. **Infinite solutions:** There are infinitely many solutions to a given differential equation due to the presence of arbitrary constants. Each distinct choice of constants results in a different solution.
2. **Number of arbitrary constants:** The solution to a differential equation of order n has n generally has n arbitrary constants. However, it is crucial to note that this rule does not always apply. For instance, the second-order equation $(y'')^2 + y^2 = 0$ [3] has only one solution $y = 0$, which is devoid of arbitrary constants.

References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.
- [2] G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd ed. Boca Raton, FL: CRC Press, 2016.
- [3] M. Tenenbaum and H. Pollard, *Ordinary Differential Equations*, reprint of the 1963 original ed. Dover Publications, 1985, dover Books on Mathematics.